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Journal:	NEJM AI
Manuscript ID	AI-25-00939
Manuscript Type:	Original Article
Date Submitted by the Author:	20-Aug-2025
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Application of Large Language Machine Learning Model to Enhance Screening Accuracy and Efficiency in Systematic Reviews and Meta-Analyses

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Short title: Application of Machine Learning in Systematic Reviews

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Abstract

Background: Manual screening in systematic reviews and meta-analyses (SRMAs) is time-consuming, error-prone, and inconsistent. We, therefore, proposed an agentic AI framework using large language models (LLMs) to improve the precision, efficiency, and interpretability during SRMAs’ manual screening process.

Methods: We developed an agentic AI framework that included four intelligent agents. The classifier agent first categorized articles based on titles and abstracts into “Potentially Relevant”, “Likely Irrelevant”, or “Uncertain”, prioritizing sensitivity to avoid premature exclusion. The detailed screening agent subsequently utilized a Chain-of-Thought (CoT) structured prompt aligned with the PICOS framework to evaluate retained citations systematically. To ensure decision integrity, the reviewer agent audited outputs and invoked the improvement agent if needed, such as when inconsistencies were detected, mimicking peer-review dynamics.

Results: We evaluated the agentic AI framework on 15 SRMAs with clearly defined quantitative outcomes and straightforward hypotheses, benchmarking against author’s screening decisions. Our agentic AI framework achieved a mean sensitivity of 0.809 (SD 0.213) and specificity of 0.885 (SD 0.120) with a false-positive rate of 0.115 (SD 0.120) and a false-negative rate of 0.191 (SD 0.213). In four SRMAs, it outperformed a recent state-of-the-art LLM-based method by improving specificity and reducing false positive rate, while maintaining a high sensitivity.

Conclusion: By combining transparency, scalability, and cost efficiency, our agentic AI framework offers a practical blueprint for applying LLMs into real-world evidence synthesis. Ongoing enhancements will aim to extend its capabilities to complex SRMAs, automate full-text screening, advance the speed and reliability, and generate the pooled estimate for SRMAs.

Introduction

Systematic reviews and meta-analyses (SRMAs) are essential to evidence-based practice, synthesizing research to inform clinical and policy decisions.^{1,2} However, as scientific literature doubles roughly every nine years, traditional SRMA methods—especially title and abstract screening—have become increasingly labor-intensive.^{3,4} Manual screening can involve thousands of citations, often requiring over 1,000 hours per SRMA and still missing 7–11% of relevant studies in single-reviewer approaches.⁴ Furthermore, Edinger & Cohen⁵ found that a substantial proportion of articles retrieved through systematic review searches were later excluded for not meeting inclusion criteria, underscoring the importance of effective early-stage filtering to reduce manual screening workload. To improve efficiency, researchers have turned to artificial intelligence (AI) and machine learning (ML), particularly natural language processing (NLP) tools, to automate relevance assessments.^{6–8} Large language models (LLMs), such as OpenAI’s GPT series, offer a significant advancement via zero-shot learning, enabling interpretation without task-specific training.⁹ Although initial studies report sensitivities of 60–90% and specificities of 10–60%, many may fail to address integration into SRMA workflows or the complexity of nuanced decision-making.^{4,10}

Building on these foundations, Dennstädt et al.¹¹ explored the use of LLMs for title and abstract screening in the biomedical domain, leveraging structured prompts to guide relevance assessments. Their findings highlighted the promise of LLMs in automating citation screening, with high sensitivity (up to 100% in some datasets) but lower specificity, necessitating further refinement for practical application. Expanding this work, Oami *et al.*^{12,13} evaluated the diagnostic accuracy of GPT-4 Turbo for title and abstract screening across five clinical questions. They reported acceptable sensitivity and reasonably high specificity, with post hoc prompt modifications further improving accuracy. Importantly, LLM-assisted screening

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significantly reduced the time required to review 100 studies compared with conventional methods, while also identifying citations missed by human reviewers. However, these studies highlighted the tendency of LLMs to generate false positives, necessitating reconciliation during the screening process. More recently, Dai *et al.*¹⁴ assessed GPT-4o and Claude-3.5 Sonnet across six thoracic surgery meta-analyses, reporting 73% sensitivity (rising to 98% with post-hoc revisions) and 99% specificity, though the generalizability of these findings beyond this thoracic surgery domain remains uncertain.

Building on these advancements and acknowledging the limitations of current evidence synthesis, our study leverages the capabilities of GPT-4 in a two-tiered framework to streamline the SRMA screening process. By incorporating structured prompts and a Chain-of-Thought reasoning framework, we aim to improve the accuracy and efficiency of LLM-assisted screening by adopting fine-tuning prompt instructions to align with the review's objectives. We hypothesized that this approach could overcome key limitations identified in prior research, such as ambiguous abstracts and the heavy burden of reconciliation, while providing a more transparent and adaptable framework for integrating LLMs into SRMA workflows.

Methods

Data Sources and Preparation

We evaluated the performance of our LLM-based screening model using data and results from 15 SRMAs. All SRMAs had a clearly articulated hypothesis based on quantitative research data, as well as featured detailed search strategies, comprehensive keywords, and registered PROSPERO numbers, and were published in reputable journals including *BMJ*, *JAMA*, and the *Lancet journals*. Detailed information for each paper is summarized in **Supplementary Table 1**. In addition, another four SRMAs were selected from a prior AI-related publication for model performance comparison purposes.

We replicated each SRMA authors' original search strategies in their stated databases to retrieve candidate articles. Our screening and filtering algorithms were then applied to these candidates, and the model's results were compared with those obtained through manual screening conducted by the study team as published. For each SRMA, we extracted the title, abstract, objectives (as mentioned in the Introduction section), and inclusion criteria (from the Methods section). This extraction approach was chosen because objectives are typically found in the final paragraph of the Introduction, or occasionally in the first paragraph of the Conclusion, while most SRMAs provide detailed selection criteria in the Methods section. By explicitly presenting the objectives and methodologies, the model is better equipped to evaluate each study against the systematic review's PICOS dimensions (Population, Intervention, Control, Outcomes, and Study design).

Design of a Multi-Agent Framework

To address the growing complexity and workload of systematic review screening, we designed a novel multi-agent framework that integrates LLM agents into a structured, autonomous evaluation pipeline. Our agentic framework entitled "Large Language Model Understanding Systematic Review & Meta-Analysis Integrated In Networked Agents" (LUMINA) comprised four intelligent agents: classifier agent, detailed screening agent, reviewer agent and improvement agent. After retrieving articles from all the databases pre-identified by the publications, the classifier agent emphasized inclusivity by rapidly filtering clearly irrelevant studies, after which the detailed screening agent conducts an in-depth PICOS-guided evaluation. To avoid LLM hallucinations and ensure reliability, reviewer agent was introduced to audit each agent's outputs and reconcile their decisions. If a decision was deemed incorrect by reviewer agent, the improvement agent would then be invoked to revise or enhance it, after which the reviewer agent re-evaluates the outcome until consensus is reached. Overall, this

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agentic framework integrated a two-tiered filtering process with an autonomous self-evaluation loop, enabling efficient large-scale screening while guaranteeing accuracy, transparency, and reproducibility, which are key priorities in evidence-based medical synthesis.

1. First-Tier: Classifier Agent

The preparatory stage of citation screening in SRMAs, keyword-based searches in literature databases, often yield a large volume of irrelevant articles. To efficiently eliminate those clearly irrelevant candidate articles while conserving computational resources, the Classifier Agent performed a rapid initial screening of citations retrieved from pre-identified databases. It was designed to filter out clearly irrelevant articles while ensuring that potentially relevant articles are not prematurely excluded. Based on the titles and abstracts of a target systematic review and the candidate studies, the Classifier Agent categorized each candidate article into one of three broad categories as follows: “*potentially relevant*”, “*likely irrelevant*” and “*uncertain*”. By design, the Classifier Agent adopted a lenient classification strategy by favouring the inclusion of articles unless they were clearly irrelevant. By restricting input to titles and abstracts, the Classifier Agent enabled rapid classification while preserving high sensitivity. Articles categorized as “*potentially relevant*” or “*uncertain*” would then be subsequently advanced to the next step (i.e., the detailed screening agent) for more rigorous evaluation.

2. Second-Tier: Detailed Screening Agent

Candidate articles that had passed through the classifier agent tend to be more homogeneous and broadly relevant to the targeted topic defined by the SRMAs. To determine which studies warrant inclusion, a more in-depth evaluation was therefore required. The Detailed Screening Agent served this purpose by delivering context-aware humanlike assessments to inform inclusion decisions. Chain-of-Thought (CoT) prompting enables LLMs to generate intermediate reasoning steps, improving performance in complex tasks by decomposing

problems into structured sequence.¹⁵ Recent work such as AlpaPICO¹⁶ demonstrated that LLMs could effectively extract PICOS elements from clinical trial documents using both in-context learning and instruction tuning.¹⁷ Building on these insights, LUMINA's Detailed Screening Agent integrated CoT reasoning with the PICOS framework, combining CoT's structured analytical approach with a standardized, widely adopted evidence-synthesis method.

The agent treated each PICOS dimension as a distinct sub-task, using one-shot guided prompts to derive filtering criteria and evaluate candidate articles. For instance, under "*Population*", it identified relevant patient groups by considering factors such as age (e.g., paediatric populations, pregnant population), gender (e.g., male, female), medical condition (e.g., hypertension, prediabetes), and clinical data (e.g., use of medication), following AlpaPICO's finding that few-shot prompting substantially outperformed zero-shot setups.¹⁶ This process was repeated across all five dimensions, producing intermediate evaluations to justify inclusion or exclusion. After completing dimension-level assessments, the agent synthesized its findings into an overall judgement, applying a leniency principle to include partially aligned but potentially valuable studies. This stepwise approach enhanced transparency, verifiability, and alignment with human expert reasoning.

3. Self-Improvement: Reviewer Agent and Improvement Agent

To enhance the reliability and accuracy of screening decisions, we implemented a Reviewer Agent and an Improvement Agent within our multi-agent framework. The Reviewer Agent operationalized the LLM-as-a-Judge paradigm,¹⁸ leveraging large language models as autonomous evaluators to audit outputs from the classifier and detailed screening agents. It systematically examined each decision and its justification, assessing whether the reasoning aligned with the evidence cited and the intended purpose of the evaluated agent. When discrepancies, methodological inconsistencies, or questionable judgments were identified, the

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Reviewer Agent generated detailed rebuttals. These rebuttals prompted the Improvement Agent to re-evaluate the same articles, incorporating the reviewer’s feedback into a refined analysis while reusing the original agent’s design. The revised output was then re-audited by the reviewer agent, and this iterative process continued until consensus was reached.

This self-improvement loop closely mirrored the revision phase in peer-reviewed systematic reviews, where reviewers request clarification or methodological refinement.¹⁹ By iteratively refining evaluations and integrating multi-level feedback, the Reviewer Agent and Improvement Agent reduced inconsistencies, improved reasoning transparency, and strengthened overall decision quality. This process ultimately enhanced the robustness and reliability of the multi-agent framework for evidence synthesis

4. Selection of Models and Decoding Settings

Given the high token volume in large-scale article screening, we adopted a cost-efficient model selection strategy. GPT-4o-mini was used for the Classifier and Detailed Screening Agents, balancing speed, cost, and context capacity for processing large text volumes. Its reasoning performance was enhanced through Chain-of-Thought (CoT) structured prompts, which improve analytical capabilities even in smaller models. For the Reviewer Agent, we selected GPT-o3-mini, which integrates embedded reasoning logic, reducing the need for lengthy CoT instructions while maintaining step-by-step evaluation (<https://platform.openai.com/docs/guides/text?api-mode=chat#choosing-a-model>). This combination provided a favourable balance between accuracy, efficiency, and cost.

To improve prompt clarity, we used structured textual markers: “###” for section headings, “_” for sublists, and “*****” to highlight key labels. These patterns served as structural cues for both human readers and LLM token parsing. While LLMs do not interpret visuals, such

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3 markers function as delimiters, helping the model segment instructions, emphasize critical
4 information, and process steps sequentially. This structured formatting complemented the CoT
5 approach, encouraging isolated reasoning for each component before synthesizing a final
6 decision.
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13 To complement aggregate performance metrics, we conducted a detailed, end-to-end
14 walk-through of LUMINA applied to a real-world systematic review, illustrating how each
15 agent contributed unique reasoning and how the self-regulatory loop improved decision
16 quality. We selected Chen *et al.*²⁰ entitled “Gestational Diabetes Mellitus and Development of
17 Intergenerational overall and Subtypes of Cardiovascular Disease: A Systematic Review and
18 Meta-Analysis”, because it had (1) clearly defined exposure and outcome, (2) well-executed
19 methodology, (3) comprehensive keywords, (4) accessible databases, and (5) publication in a
20 reputable journal. Importantly, it was led by the same principal investigator (L.-J.L.) as the
21 present study, ensuring continuity and reliability. The search process would therefore be guided
22 by established expertise and prior experience. Candidate articles were first retrieved using the
23 keywords and database searches reported by Chen *et al.* (**Table 1**), with only titles and abstracts
24 provided to the classifier agent alongside the review’s title and abstract for initial
25 categorization.
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44 Articles advancing past this stage were evaluated by the detailed screening agent. To improve
45 alignment with the original review framework, we supplemented the agent’s input with the
46 final paragraph of Chen *et al.*’s introduction (capturing the study objective) and the inclusion
47 criteria from its methods section. This context-enriched input allowed the agent to assess
48 candidate studies against Chen *et al.*’s predefined criteria with greater precision.
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57 To illustrate the functionality of each agent in the LUMINA framework, we selected three
58 representative candidate articles as examples, to demonstrate the following: (1) How an article
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entitled “Evaluation of fetal myocardial performance index in gestational diabetes mellitus” was filtered out by the Classifier Agent due to irrelevance (**Annex 1**); (2) How an article entitled “Association Between Gestational Diabetes Mellitus and the Risks of Type-Specific Cardiovascular Diseases” was successfully passed all agents in LUMINA framework (**Annex 2**); and (3) How an article entitled “Pregnancy Complications and Risk of Cardiovascular Disease Later in Life: A Nationwide Cohort Study” where a disagreement arose between the Detailed Screening Agen and the Reviewer Agent, and triggered the self-improvement loop using Improvement Agent (**Annex 3**).

Statistical Analysis

We evaluated the performance of the LUMINA framework by comparing its decision accuracy against the gold standard, defined as the original papers included and reported in each SRMA. For each SRMA, standard article-level metrics were calculated: sensitivity, specificity, false-positive rate (FPR), false-negative rate (FNR), positive predictive value (PPV), and negative predictive value (NPV). Descriptive statistics summarized these metrics across all 15 SRMAs, reporting means, standard deviations (SD), quartile (25th, 50th, 75th) and interquartile range (IQR) to characterize central tendency and variability. Metrics were computed individually per SRMA and aggregated to estimate overall model effectiveness. To further validate the performance of our multi-agent framework, we implemented and tested approaches from Li et al.,²¹ and Strachan J.A.²² on the same 15 SRMAs. Moreover, we conducted a comparative experiment based on Tran *et al.*⁴, who used five handcrafted prompt templates from PICOS dimensions with manual fine-tuned filtering criteria embedded into each. Their method queried an LLM iteratively across PICOS dimensions to yield inclusion decisions. All analyses were performed by R package (version 4.4.2). The student’s t-test was applied in two independent groups comparison while One-way ANOVA was applied in multiple independent groups comparison. A two-sided p value at 0.05 was considered significant.

To evaluate efficiency gains, we recorded and analyzed the screening time and estimated associated costs of the automated process relative to manual screening, using metrics such as total screening time per SRMA, articles screened per hour, and overall resource utilization.

Results

The selected SRMAs, along with detailed results, are summarized in **Table 1**. **Table 2** shows that the overall performance of LUMINA has demonstrated high performance: mean FPR as 0.115 (SD 0.12, ranging from 0.003 to 0.377), mean FNR as 0.191 (SD 0.213, ranging from 0 to 0.692), mean sensitivity as 0.809 (SD 0.213, ranging from 0.308 to 1.000), and mean specificity as 0.885 (SD 0.120, ranging from 0.623 to 0.991). Notably, the average NPV approached 100% (mean 0.994, SD 0.009, ranging from 0.958 to 1.000), highlighting excellent reliability in excluding irrelevant studies. However, the mean PPV was low as 0.169 (SD 0.108, ranging from 0.027 to 0.444), likely reflecting a high false positive count due to the intentionally lenient threshold aimed at minimizing false negatives.

To further evaluate LUMINA's performance, we implemented and tested approaches from Li *et al.*,²¹ and Strachan J.A.²² on the same 15 SRMAs (**Supplementary Tables 2 & 3**). LUMINA outperformed all of them, while also revealing their respective limitations. Li *et al.*,²¹ evaluated the effectiveness of LLM in abstract screening while Strachan J.A.²² applied prompts for paper inclusion. Compared across LUMINA, Li *et al.*,²¹ and Strachan J.A.²² methods, our LUMINA framework exhibited significant performance merits, including the highest FPR (0.115 [0.120] vs. 0.020 [0.035] vs. 0.058 [0.086], $p=0.03$), the lowest FNR (0.191 [0.213] vs. 0.630 [0.288] vs. 0.421 [0.303], $p<0.0001$), the highest sensitivity (0.809 [0.213] vs. 0.370 [0.288] vs. 0.579 [0.303], $p<0.0001$). The NPV and PPV were comparable across the three methods (**Supplementary Table 4 & Figure 2A**).

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Our LUMINA framework also evaluated the performance accuracy against GPT-3.5 turbo by Tran *et al.*⁴ (**Table 3**), and exhibited an improved performance in all metrics, among which FNR (0.00 [0.00] vs. 0.138 [0.068], $p=0.08$) and sensitivity (1.00 [0.00] vs. 0.863 [0.068], $p=0.008$) reached significance (**Supplementary Table 5 and Figure 2B**).

In terms of efficiency, LUMINA screened 10 articles in approximately 6.67 minutes (400 seconds) at a total computational cost of USD 0.07. These results demonstrate the practical advantages of our framework in reducing missed studies and minimizing manual workload, thereby enhancing the efficiency and methodological rigor of systematic reviews.

Discussion

In LUMINA, we developed an LLM-based multi-agent framework that integrated advanced human-like reasoning and a self-evaluation loop, enabling a more efficient and systematic approach to literature screening and study selection in SRMAs. We tested LUMINA across a set of 15 pre-identified SRMAs, each characterized as a quantitative study with a structured hypothesis, featuring a single exposure and a single outcome. In our experiment, LUMINA reduced the size of the screening candidate pool by an order of magnitude while maintaining a high sensitivity rate. Collectively, our comparative results highlighted the practical benefits of the multi-agent LLM framework, which effectively minimized the risk of missing critical studies and significantly lowered the burden of manual review, improving both efficiency and methodological rigor in SRMAs.

Our agent showed substantial improvements in specificity, significantly reducing false positives compared to Tran *et al.*⁴'s method. For instance, in the Kiesswetter *et al.*²³ SRMA, LUMINA achieved a high specificity (0.952) and a low FPR (0.048), whereas Tran *et al.*'s best-performing model only reached moderate specificity (0.558) with a very high FPR (0.442). Similarly, in the Sbidian *et al.*²⁴ SRMA, LUMINA attained a specificity of 0.628 with a moderate FPR (0.372), outperforming Tran *et al.*'s specificity of 0.243 with a very high FPR

(0.757). Higher specificity in such complex settings directly reduces screening burden, demonstrating LUMINA's capacity to streamline SRMA workflows. Notably, our model maintained perfect sensitivity (1.00) across all four SRMAs, while Tran et al.'s sensitivity ranged from 0.817 to 0.998. This consistency is crucial, as missing relevant articles can substantially compromise the validity and reliability of systematic reviews.²⁵

To further evaluate LUMINA's performance, we benchmarked it against the approaches proposed by Li *et al.*,²¹ and Strachan J.A.²² using the same 15 SRMAs. LUMINA consistently outperformed both methods, while also revealing their inherent limitations. For instance, Li *et al.*,²¹ used sets of manually predefined filtering criteria embedded into their prompt templates. In their design, the LLM answered each criterion with a simple "yes" or "no," followed by an overall explanation—an approach that introduced ambiguity in how the criteria were actually evaluated. Similarly, Strahan J.A.²² also relied on predefined criteria, asking the LLM to identify PICO themes with binary responses. However, the effectiveness of such methods was highly contingent upon the quality of the filtering criteria, which required substantial manual efforts and time to optimize for each individual SRMA.

Prompt design is central to integrating LLMs into SRMA pipelines. Lieberum *et al.*²⁶ suggested that slight variations in the prompts used could affect the LLM's output significantly, yet many medical research team lack prompt engineering expertise. Zaghir *et al.*²⁷ found that among 114 recent LLM-based medical studies, 64% used no benchmark model when crafting their prompts, which is in contrary to best practices. LUMINA has addressed these shortcomings through structured, interpretable prompts based on the CoT and PICOS frameworks, thereby reducing engineering complexity while improving reliability.

Finally, hallucination remains a core limitation of LLMs. As mentioned by Chiang and Roberts,²⁸ systems must be designed to accommodate rather than eliminate this risk. LUMINA

incorporates a self-evaluation mechanism to emulate the analytical reasoning of human reviewers and mitigate hallucination. This feature mirrors peer review in traditional SRMAs, enhancing methodological rigor and quality control.²⁹

Strengths and Limitations

The key strength of our study is the development of a high-performing LLM-based agentic framework – LUMINA, that automates SRMA citation screening, guaranteeing efficiency and precision. In addition, we evaluated LUMINA across 15 SRMAs from diverse medical fields and conducted comprehensive comparison analyses against the LLM method by Trans *et al.*⁴ Through rigorous study design and benchmarking, we demonstrated the robustness and superior performance of our agentic framework. However, we acknowledge that our study has several limitations. Firstly, currently LUMINA works well on quantitative SRMAs with straightforward hypothesis—clearly defined one exposure and one outcome. The generalizability and robustness of LUMINA on more complex SRMAs remains to be improved further. Moving forward, we will enhance LUMINA’s capability to handle SRMAs that use quantitative methods involving multiple exposures and outcomes, as well as qualitative SRMAs where exposures and outcomes may be presented in more layered or complex forms. Secondly, our pre-identified SRMAs are published and available online, thus they could be used as corpuses in LLMs’ we used when they are being trained. Therefore, we plan to collaborate with SRMA groups to test the model in the actual SRMA process, thereby eliminating inherent bias and directly verifying the practicality of our model in real-world SRMA processes. Thirdly, LUMINA’s performance was evaluated solely against the screening and selection decisions of each SRMA study team, which inherently involve some subjectivity and may not constitute an absolute gold standard. Last but not least, given the rapid evolution of LLMs, ongoing monitoring will be necessary to ensure the reproducibility of LUMINA’s results.

Future direction

Besides improving LUMINA's performance on more complex SRMAs, our work will also focus on extending the framework to exploit LLM's potential in other stages of the SRMA process, such as keywords generation and AI-conducted meta-analysis. Aside from addressing the limitations mentioned above, there are a few more research interests that are worthwhile exploring in the future. Aside from OpenAI's GPT series, our model could be applicable with other LLMs available, such as Claude and Deepseek. These models are trained differently and could potentially yield better performance.

In summary, LUMINA showcases the promise of agentic AI in automating SRMA literature screening through a team of autonomous, reasoning agents. Using structured CoT logic and the PICOS framework, it mirrors human review while ensuring transparency and reliability. Its self-evaluating agents enable dynamic refinement, fostering trust and robustness. Most importantly, LUMINA's modular, adaptive design lays the foundation for future AI systems that integrate seamlessly into clinical workflows and scale across evidence synthesis domains.

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Conflict of Interest Disclosures: No financial disclosure, and the other authors also have no financial disclosures relevant to this article. No financial or non-financial benefits have been received or will be received from any party related directly or indirectly to the individuals of this article.

Data Sharing Statement: Data will be available upon research agreement by contacting the corresponding authors of the study team.

Funding/Support: This project is funded by Artificial Intelligence in Medicine Grant Call (AliM08052024), National University of Singapore, Singapore.

Role of the Funder/Sponsor: The funders/sponsors had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

References

1. Cook DJ, Mulrow CD, Haynes RB. Systematic reviews: synthesis of best evidence for clinical decisions. *Ann Intern Med* 1997;126(5):376-80. (In eng). DOI: 10.7326/0003-4819-126-5-199703010-00006.
2. Wang XM, Zhang XR, Li ZH, Zhong WF, Yang P, Mao C. A brief introduction of meta-analyses in clinical practice and research. *J Gene Med* 2021;23(5):e3312. (In eng). DOI: 10.1002/jgm.3312.
3. Gates A, Johnson C, Hartling L. Technology-assisted title and abstract screening for systematic reviews: a retrospective evaluation of the Abstrackr machine learning tool. *Systematic Reviews* 2018;7(1):45. DOI: 10.1186/s13643-018-0707-8.
4. Tran VT, Gartlehner G, Yaacoub S, et al. Sensitivity and Specificity of Using GPT-3.5 Turbo Models for Title and Abstract Screening in Systematic Reviews and Meta-analyses. *Ann Intern Med* 2024;177(6):791-799. DOI: 10.7326/M23-3389.
5. Edinger T, Cohen AM. A large-scale analysis of the reasons given for excluding articles that are retrieved by literature search during systematic review. *AMIA Annu Symp Proc* 2013;2013:379-87. (<https://www.ncbi.nlm.nih.gov/pubmed/24551345>).
6. Marshall IJ, Wallace BC. Toward systematic review automation: a practical guide to using machine learning tools in research synthesis. *Systematic Reviews* 2019;8(1):163. DOI: 10.1186/s13643-019-1074-9.
7. Sanghera R, Thirunavukarasu AJ, El Khoury M, et al. High-performance automated abstract screening with large language model ensembles. (<https://ui.adsabs.harvard.edu/abs/2024arXiv241102451S>).
8. Guo E, Gupta M, Deng J, Park Y-J, Paget M, Naugler C. Automated Paper Screening for Clinical Reviews Using Large Language Models: Data Analysis Study. *J Med Internet Res* 2024;26:e48996. (Original Paper) (In English). DOI: 10.2196/48996.
9. Wang S, Scells H, Zhuang S, Potthast M, Koopman B, Zuccon G. Zero-shot Generative Large Language Models for Systematic Review Screening Automation. (<https://ui.adsabs.harvard.edu/abs/2024arXiv240106320W>).
10. Syriani E, David I, Kumar G. Assessing the Ability of ChatGPT to Screen Articles for Systematic Reviews. (<https://ui.adsabs.harvard.edu/abs/2023arXiv230706464S>).
11. Dennstädt F, Zink J, Putora PM, Hastings J, Cihoric N. Title and abstract screening for literature reviews using large language models: an exploratory study in the biomedical domain. *Systematic Reviews* 2024;13(1):158. DOI: 10.1186/s13643-024-02575-4.
12. Oami T, Okada Y, Nakada T-a. Performance of a Large Language Model in Screening Citations. *JAMA Network Open* 2024;7(7):e2420496-e2420496. DOI: 10.1001/jamanetworkopen.2024.20496.
13. Oami T, Okada Y, Nakada T-a. GPT-3.5 Turbo and GPT-4 Turbo in Title and Abstract Screening for Systematic Reviews. *JMIR Med Inform* 2025;13:e64682. DOI: 10.2196/64682.
14. Dai Z-Y, Wang F-Q, Shen C, et al. Accuracy of Large Language Models for Literature Screening in Thoracic Surgery: Diagnostic Study. *J Med Internet Res* 2025;27:e67488. (Original Paper) (In English). DOI: 10.2196/67488.
15. Wei J, Wang X, Schuurmans D, et al. Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. . *Computer Science--Computation and Language* 2022 (<https://arxiv.org/abs/2201.11903>).
16. Ghosh M, Mukherjee S, Gnaguly A, Basuchowdhuri P, Naskar SK, Ganguly D. AlpaPICO: Extraction of PICO Frames from Clinical Trial Documents Using LLMs.

Computer Science--Computation and Language 2024 (<https://arxiv.org/abs/2409.09704>).

17. Eriksen MB, Frandsen TF. The impact of patient, intervention, comparison, outcome (PICO) as a search strategy tool on literature search quality: a systematic review. *J Med Libr Assoc* 2018;106(4):420-431. DOI: 10.5195/jmla.2018.345.

18. Gu J, Jiang X, Shi Z, et al. A Survey on LLM-as-a-Judge. *Computer Science--Computation and Language 2024* (<https://arxiv.org/abs/2411.15594>).

19. Baxter L. How to Peer Review a Systematic Review: A Peer-Reviewer's Guide to Reviewing Reviews. *J Clin Pharmacol* 2025. DOI: 10.1002/jcph.70036.

20. Chen A, Tan B, Du R, et al. Gestational diabetes mellitus and development of intergenerational overall and subtypes of cardiovascular diseases: a systematic review and meta-analysis. *Cardiovasc Diabetol* 2024;23(1):320. DOI: 10.1186/s12933-024-02416-7.

21. Li M, Sun J, Tan X. Evaluating the effectiveness of large language models in abstract screening: a comparative analysis. *Syst Rev* 2024;13(1):219. DOI: 10.1186/s13643-024-02609-x.

22. Strachan JA. Designing GPT3 prompts to screen articles for systematic reviews of RCTs. *Health Policy and Technology* 2025;14(1).

23. Kiesswetter E, Stadelmaier J, Petropoulou M, et al. Effects of Dairy Intake on Markers of Cardiometabolic Health in Adults: A Systematic Review with Network Meta-Analysis. *Adv Nutr* 2023;14(3):438-450. DOI: 10.1016/j.advnut.2023.03.004.

24. Sbidian E, Chaimani A, Garcia-Doval I, et al. Systemic pharmacological treatments for chronic plaque psoriasis: a network meta-analysis. *Cochrane Database Syst Rev* 2022;5(5):CD011535. DOI: 10.1002/14651858.CD011535.pub5.

25. Petticrew M, Roberts H. Systematic reviews--do they 'work' in informing decision-making around health inequalities? *Health Econ Policy Law* 2008;3(Pt 2):197-211. DOI: 10.1017/S1744133108004453.

26. Lieberum J-L, Töws M, Metzendorf M-I, et al. Large language models for conducting systematic reviews: on the rise, but not yet ready for use—a scoping review. *Journal of Clinical Epidemiology* 2025;181:111746. DOI: <https://doi.org/10.1016/j.jclinepi.2025.111746>.

27. Zaghir J, Naguib M, Bjelogrić M, Névél A, Tannier X, Lovis C. Prompt engineering paradigms for medical applications: scoping review and recommendations for better practices. (<https://ui.adsabs.harvard.edu/abs/2024arXiv240501249Z>).

28. Banerjee S, Agarwal A, ingla S. LLMs will always hallucinate, and we need to live with this. . *Statistics--Machine Learning* 2024. DOI: <https://doi.org/10.48550/arXiv.2409.05746>.

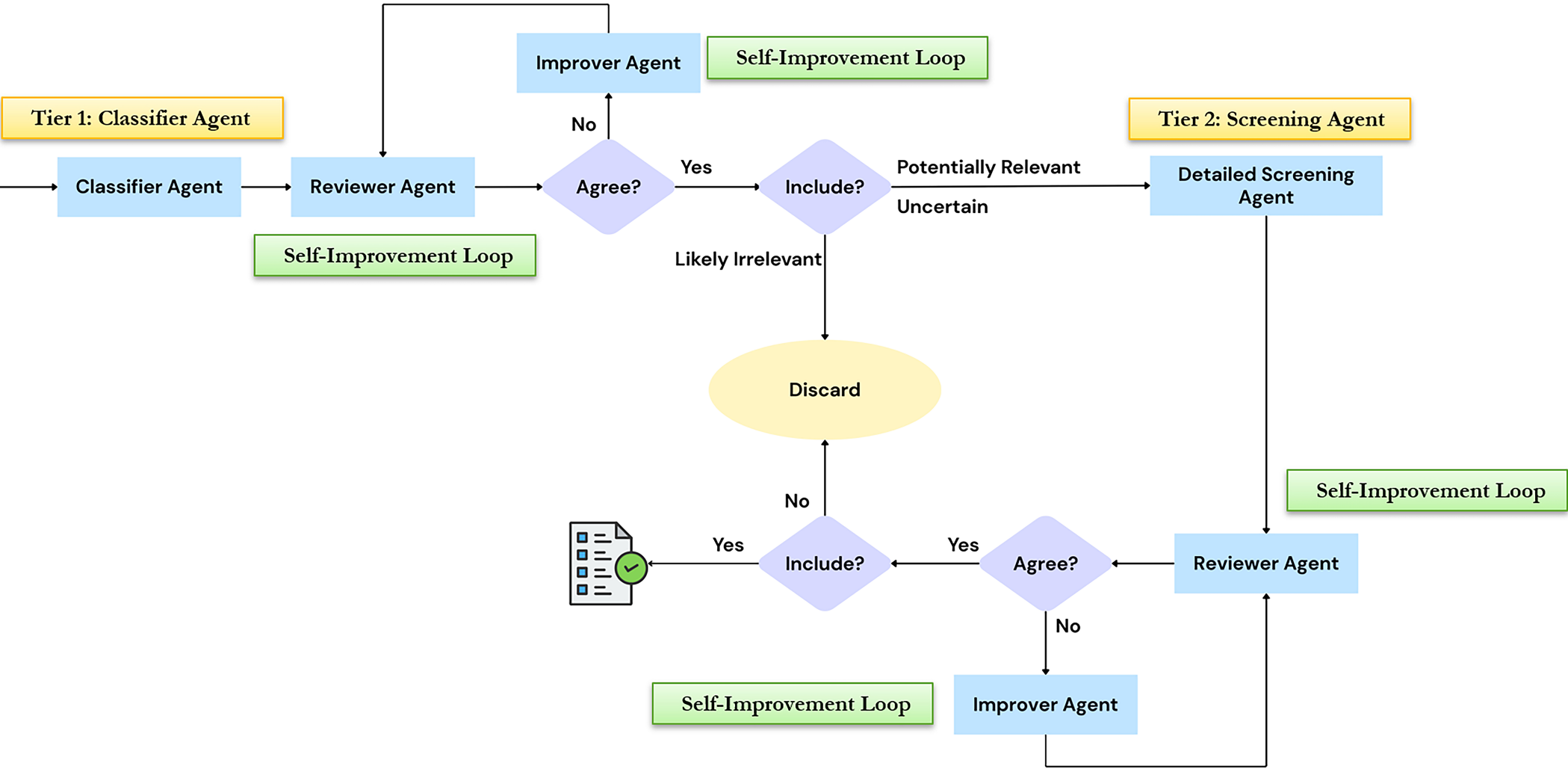
29. Kelly J, Sadeghieh T, Adeli K. Peer Review in Scientific Publications: Benefits, Critiques, & A Survival Guide. *EJIFCC* 2014;25(3):227-43. (<https://www.ncbi.nlm.nih.gov/pubmed/27683470>).

Figure legends

Figure 1. Workflow diagram of our multi-agent framework design

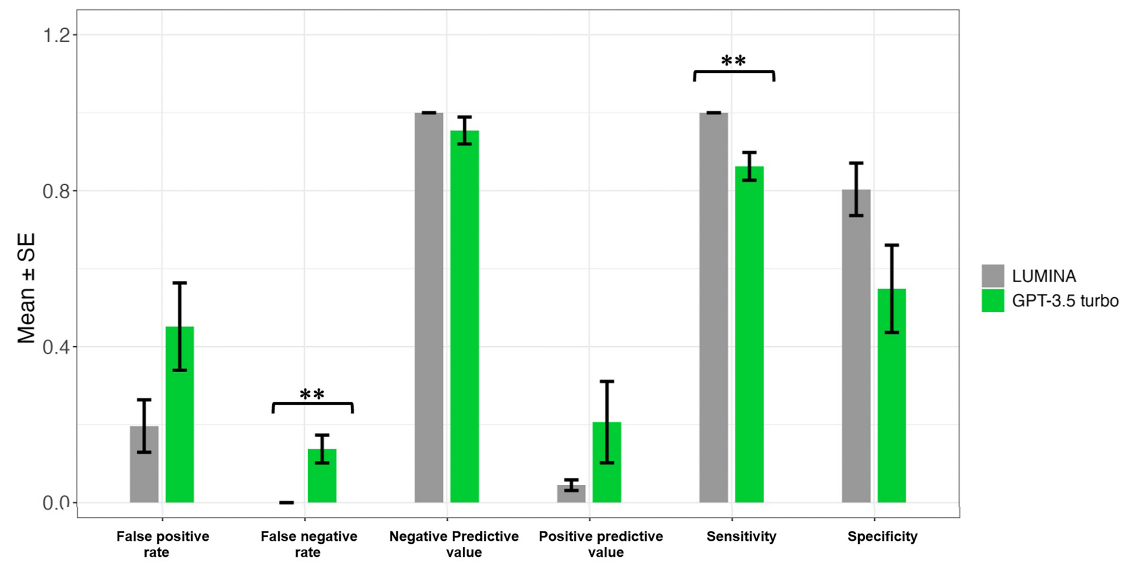
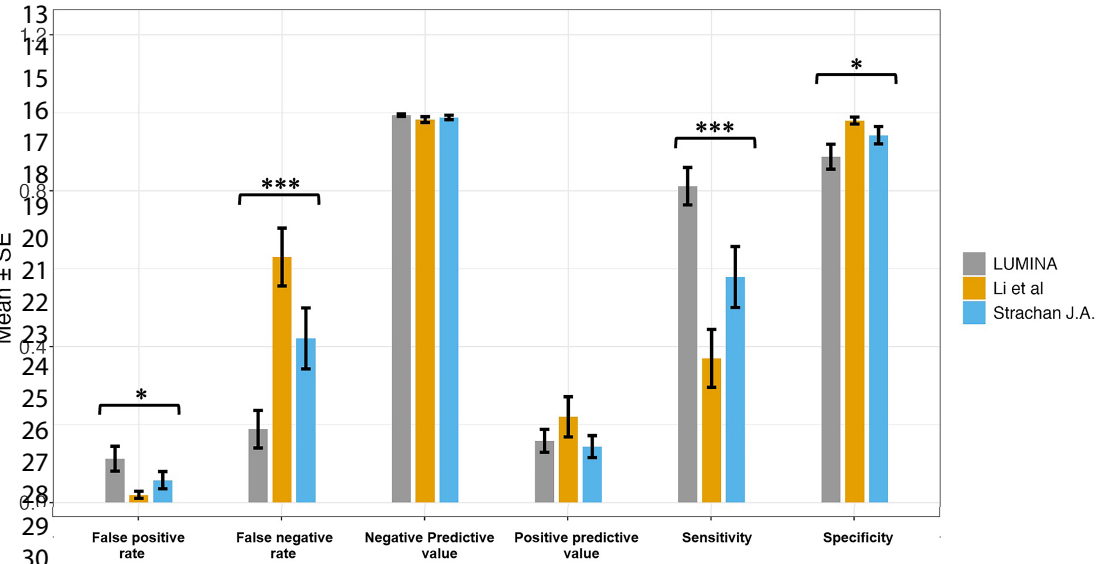
Figure 2. Comparison of performance metrics between LUMINA framework with other published machine learning methods in systematic review and meta-analysis (SRMA) screening. Figure 2A represents the performance metrics comparison between LUMINA framework and Li *et al.*²¹ and Strachan J.A.²² published methods across the 15 SRMAs selected in this study; Figure 2B represents the performance metrics comparison between LUMINA framework and GPT-3.6 turbo based on the four SRMAs selected by Tran *et al.*⁴

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Table 1. Performance of LUMINA framework in 15 selected systematic reviews and meta-analyses, against the originally included publications.

SRMA Reference s According to Table 1 sequence	Actual paper included	Total paper included by LUMINA	Papers classified as true positive (TP)	Papers classified as true negative (TN)	Papers classified as false positive (FP)	Papers classified as false negative (FN)	False positive rate (FPR) FP/(FP+TN)	False negative rate (FNR) FN/(FN+TP)	Negative predictive value (NPV) TN/(TN+FN)	Positive predictive value (PPV) TP/(TP+FP)	Sensitivity TP/(TP+FN)	Specificity TN/(TN+FP)
1	32	12961	39	12241	666	15	0.052	0.278	0.999	0.055	0.722	0.948
2	12	8526	18	8425	78	5	0.009	0.217	0.999	0.188	0.783	0.991
3	113	5793	90	3551	2148	4	0.377	0.043	0.999	0.040	0.957	0.623
4	45	11656	42	11387	224	3	0.019	0.067	0.999	0.158	0.933	0.981
5	9	2424	11	2259	152	2	0.063	0.154	0.999	0.068	0.846	0.937
6	16	1286	22	1179	81	4	0.064	0.154	0.997	0.214	0.846	0.936
7	15	1898	26	1642	227	3	0.122	0.103	0.998	0.103	0.897	0.879
8	13	3444	14	2921	509	0	0.148	0.000	1.000	0.027	1.000	0.852
9	14	583	14	467	101	1	0.178	0.067	0.998	0.122	0.933	0.822
10	10	2564	13	2461	90	0	0.035	0.000	1.000	0.126	1.000	0.965
11	10	164	15	113	31	5	0.215	0.250	0.958	0.326	0.750	0.785
12	28	8128	31	7989	95	13	0.012	0.296	0.998	0.246	0.705	0.988
13	27	11938	13	11776	139	10	0.012	0.435	0.999	0.086	0.565	0.988
14	25	328	4	310	5	9	0.016	0.692	0.972	0.444	0.308	0.984
15	111	1828	132	1067	619	10	0.367	0.070	0.991	0.176	0.930	0.633

Table 2. Overall performance of LUMINA across 15 systematic reviews and meta-analyses

Performance metrics	mean	Standard Deviation (SD)	Standard Error (SE)	minimum	25th percentile	50 th centile	75 th percentile	maximum	Interquartile range (IQR)
False positive rate (FPR)	0.115	0.120	0.031	0.009	0.016	0.063	0.148	0.377	0.133
False negative rate (FNR)	0.191	0.213	0.055	0.0000	0.067	0.154	0.250	0.692	0.183
Negative predictive value (NPV)	0.994	0.009	0.002	0.958	0.998	0.999	0.999	1.000	0.001
Positive predictive value (PPV)	0.169	0.108	0.028	0.027	0.086	0.126	0.214	0.444	0.128
Sensitivity	0.809	0.213	0.055	0.308	0.750	0.846	0.933	1.000	0.183
Specificity	0.885	0.120	0.031	0.623	0.852	0.937	0.984	0.991	0.133

Table 3. Comparisons between GPT-3.5 turbo methods and LUMINA across four other systematic reviews and meta-analyses.

SRMA references	Methods comparisons	No of papers as false positive (FP)	No of papers as false negative (FN)	No of papers as True positive (TP)	No of papers as True negative (TN)	Total papers	False positive rate (FPR)	False negative rate (FNR)	Negative predictive value (NPV)	Positive predictive value (PPV)	Specificity	Sensitivity
Sommer	LUMINA	513	0	22	1983	2518	0.2056	0.0000	1.0000	0.0411	0.7944	1.0000
	GPT-3.5 turbo	187	44	189	252	672	0.4260	0.1888	0.8514	0.5027	0.5740	0.8112
Sommer (Version 2)	LUMINA	870	0	11	4579	5460	0.1597	0.0000	1.0000	0.0125	0.8403	1.0000
	GPT-3.5 turbo	776	22	98	3181	4077	0.1962	0.1833	0.9931	0.1122	0.8038	0.8167
Kiesswetter	LUMINA	328	0	16	6480	6824	0.0482	0.0000	1.0000	0.0465	0.9518	1.0000
	GPT-3.5 turbo	2828	11	66	3573	6478	0.4417	0.1429	0.9969	0.0228	0.5583	0.8571
Sbidian	LUMINA	1941	0	167	3272	5380	0.3724	0.0000	1.0000	0.0792	0.6276	1.0000
	GPT-3.5 turbo	3215	27	747	1115	5104	0.7423	0.0349	0.9764	0.1885	0.2577	0.9651

SRMA references:

1. SommerI, Dobrescu A, Ledinger D, et al. Outpatient treatment of confirmed COVID-19: a living, rapid review for the American College of Physicians. Ann Intern Med. 2023;176:92-104. [PMID: 36442056]doi:10.7326/M22-2202 24.
2. Sommer I, Ledinger D, Thaler K, et al. Outpatient treatment of confirmed COVID-19: a living, rapid evidence review for the American College of Physicians (Version 2). Ann Intern Med. 2023;176:1377-1385. [PMID: 37722115] doi:10.7326/M23-1626 25.
3. Kiesswetter E, Stadelmaier J, Petropoulou M, et al. Effects of dairy intake on markers of cardiometabolic health in adults: a systematic review with network meta-analysis. Adv Nutr. 2023;14:438 450. [PMID: 36914032]doi:10.1016/j.advnut.2023.03.004
4. Sbidian E, Chaimani A, Guelimi R, et al. Systemic pharmacological treatments for chronic plaque psoriasis: a network meta-analysis. Cochrane Database Syst Rev. 2023;7:CD011535.